Load the dataset

In [1]:
```
import pandas as pd
df = pd.read_csv("obesity-dataset.csv")
```

In [2]:
```
df.head()
```

Out[2]:

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0	Some
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0	Some
2	Male	23.0	1.80	77.0	yes	no	2.0	3.0	Some
3	Male	27.0	1.80	87.0	no	no	3.0	3.0	Some
4	Male	22.0	1.78	89.8	no	no	2.0	1.0	Some

Shape of the original dataset

In [3]:
```
df.shape
```

Out[3]:
```
(2111, 17)
```

Number of duplicate rows

In [4]:
```
df.duplicated().sum()
```

Out[4]:
```
21
```

Remove duplicate rows

In [5]:
```
df = df.drop_duplicates()
```

In [6]:
```
df.shape
```

Out[6]:
```
(2090, 17)
```

Data type of each column

In [7]:
```
df.dtypes
```

Out[7]:
```
Gender                            object
Age                              float64
Height                           float64
Weight                           float64
family_history_with_overweight    object
FAVC                              object
FCVC                             float64
NCP                              float64
CAEC                              object
SMOKE                             object
CH2O                             float64
SCC                               object
FAF                              float64
TUE                              float64
CALC                              object
MTRANS                            object
NObeyesdad                        object
dtype: object
```

Categories of the "Gender" column

In [8]:
```
df["Gender"].value_counts()
```

Out[8]:
```
Gender
Male      1054
Female    1036
Name: count, dtype: int64
```

Categories of the "family_history_with_overweight" column

In [9]:
```
df["family_history_with_overweight"].value_counts()
```

Out[9]:
```
family_history_with_overweight
yes    1722
no      368
Name: count, dtype: int64
```

Categories of the "FAVC" column

In [10]:
```
df["FAVC"].value_counts()
```

Out[10]:
```
FAVC
yes    1847
no      243
Name: count, dtype: int64
```

Categories of the "SMOKE" column

In [11]:
```
df["SMOKE"].value_counts()
```

Out[11]:
```
SMOKE
no     2046
yes      44
Name: count, dtype: int64
```

Categories of the "SCC" column

In [12]:
```
df["SCC"].value_counts()
```

Out[12]:
```
SCC
no     1994
yes      96
Name: count, dtype: int64
```

Label encoding "Gender", "family_history_with_overweight", "FAVC", "SMOKE" and "SCC" columns

In [13]:
```
from sklearn.preprocessing import LabelEncoder

encoders = {}
cols_to_encode = ["Gender", "family_history_with_overweight", "FAVC", "SMOKE", "SCC

for col in cols_to_encode:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoders[col] = le
```

Categories of the "CAEC" column

In [14]:
```
df["CAEC"].value_counts()
```

Out[14]:
```
CAEC
Sometimes     1761
Frequently     237
Always          53
no              39
Name: count, dtype: int64
```

Categories of the "CALC" column

In [15]:
```
df["CALC"].value_counts()
```

Out[15]:
```
CALC
Sometimes     1383
no             636
Frequently      70
Always           1
Name: count, dtype: int64
```

Manual encoding "CAEC" and "CALC" columns

In [16]:
```
frequency_mapping = {
    'no': 0,
    'Sometimes': 1,
    'Frequently': 2,
    'Always': 3
}

for col in ['CAEC', 'CALC']:
    df[col] = df[col].map(frequency_mapping)
```

One-hot encoding the "MTRANS" column

In [17]:
```
df["MTRANS"].value_counts()
```

Out[17]:
```
MTRANS
Public_Transportation    1561
Automobile                456
Walking                    55
Motorbike                  11
Bike                        7
Name: count, dtype: int64
```

In [18]:
```
df = pd.get_dummies(df, columns=['MTRANS'], dtype=int)
```

Manual encoding the "NObeyesdad" column

In [19]:
```
df["NObeyesdad"].value_counts()
```

Out[19]:
```
NObeyesdad
Obesity_Type_I         351
Obesity_Type_III       324
Obesity_Type_II        297
Overweight_Level_II    290
Normal_Weight          282
Overweight_Level_I     278
Insufficient_Weight    268
Name: count, dtype: int64
```

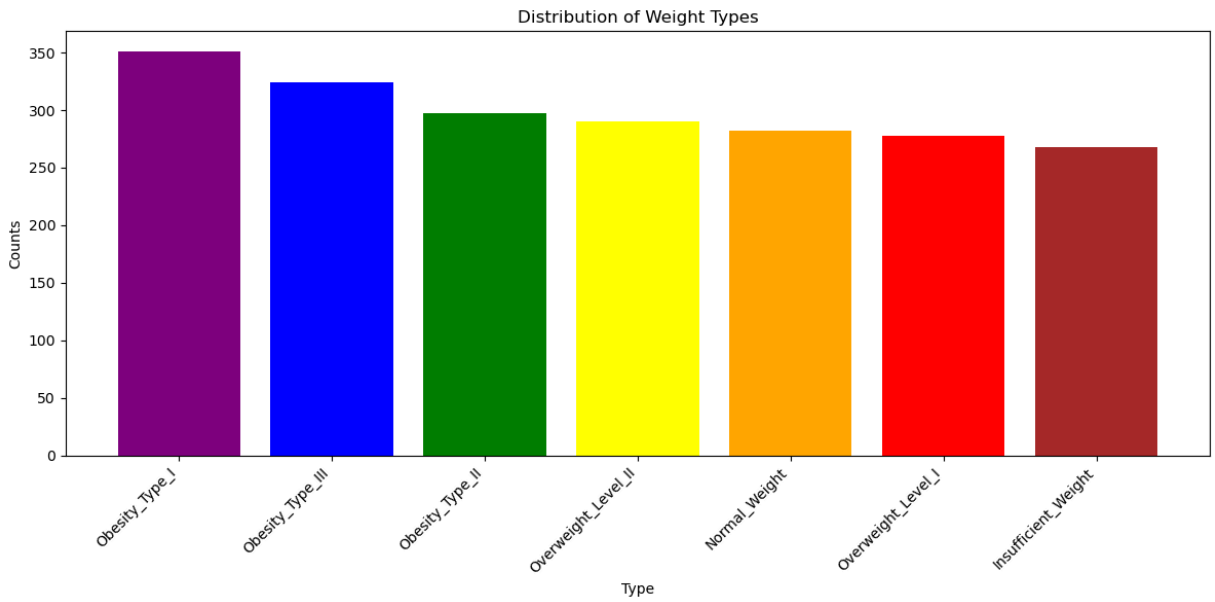
In [20]:
```
NObeyesdad_mapping = {
    "Obesity_Type_I": 0,
    "Obesity_Type_III": 1,
    "Obesity_Type_II": 2,
    "Overweight_Level_II": 3,
    "Normal_Weight": 4,
    "Overweight_Level_I": 5,
    "Insufficient_Weight": 6
}
df['NObeyesdad'] = df['NObeyesdad'].map(NObeyesdad_mapping)
```

In [21]:
```
import matplotlib.pyplot as plt

types = [
    "Obesity_Type_I",
    "Obesity_Type_III",
    "Obesity_Type_II",
    "Overweight_Level_II",
    "Normal_Weight",
    "Overweight_Level_I",
    "Insufficient_Weight"
]

counts = [351, 324, 297, 290, 282, 278, 268]

plt.figure(figsize=(12, 6))
plt.bar(types, counts, color=["purple", "blue", "green", "yellow", "orange", "red",
plt.xlabel("Type")
plt.ylabel("Counts")
plt.title("Distribution of Weight Types")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Final data type of each column

In [22]:
```
df.dtypes
```

Out[22]:
```
Gender                            int32
Age                             float64
Height                          float64
Weight                          float64
family_history_with_overweight    int32
FAVC                              int32
FCVC                            float64
NCP                             float64
CAEC                              int64
SMOKE                             int32
CH2O                            float64
SCC                               int32
FAF                             float64
TUE                             float64
CALC                              int64
NObeyesdad                        int64
MTRANS_Automobile                 int32
MTRANS_Bike                       int32
MTRANS_Motorbike                  int32
MTRANS_Public_Transportation      int32
MTRANS_Walking                    int32
dtype: object
```

Count of NULL values for each column

In [23]:
```
df.isnull().sum()
```

Out[23]:
```
Gender                              0
Age                                30
Height                             45
Weight                             55
family_history_with_overweight      0
FAVC                                0
FCVC                               40
NCP                                 0
CAEC                                0
SMOKE                               0
CH2O                                0
SCC                                 0
FAF                                60
TUE                                 0
CALC                                0
NObeyesdad                          0
MTRANS_Automobile                   0
MTRANS_Bike                         0
MTRANS_Motorbike                    0
MTRANS_Public_Transportation        0
MTRANS_Walking                      0
dtype: int64
```

Define input and target variables

In [24]:
```
X = df.drop("NObeyesdad", axis=1)
y = df["NObeyesdad"]
```

Split the data

In [25]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

Initialize the KNN Imputer

In [26]:
```
from sklearn.impute import KNNImputer
import pandas as pd

knn_imputer = KNNImputer(n_neighbors=5)
```

Fit and Transform the training data

In [27]:
```
X_train_imputed = knn_imputer.fit_transform(X_train)
```

Transform the test data

In [28]:
```
X_test_imputed = knn_imputer.transform(X_test)
```

Convert imputed data back to data frames

In [29]:
```
X_train_imputed = pd.DataFrame(X_train_imputed, columns=X_train.columns)
X_test_imputed = pd.DataFrame(X_test_imputed, columns=X_test.columns)
```

Final NULL value check for the input variables

In [30]:
```
X_train_imputed.isnull().sum()
```

Out[30]:
```
Gender                            0
Age                               0
Height                            0
Weight                            0
family_history_with_overweight    0
FAVC                              0
FCVC                              0
NCP                               0
CAEC                              0
SMOKE                             0
CH2O                              0
SCC                               0
FAF                               0
TUE                               0
CALC                              0
MTRANS_Automobile                 0
MTRANS_Bike                       0
MTRANS_Motorbike                  0
MTRANS_Public_Transportation      0
MTRANS_Walking                    0
dtype: int64
```

In [31]:
```
X_test_imputed.isnull().sum()
```

Out[31]:
```
Gender                            0
Age                               0
Height                            0
Weight                            0
family_history_with_overweight    0
FAVC                              0
FCVC                              0
NCP                               0
CAEC                              0
SMOKE                             0
CH2O                              0
SCC                               0
FAF                               0
TUE                               0
CALC                              0
MTRANS_Automobile                 0
MTRANS_Bike                       0
MTRANS_Motorbike                  0
MTRANS_Public_Transportation      0
MTRANS_Walking                    0
dtype: int64
```

In [32]:
```
df.head()
```

Out[32]:

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC
0	0	21.0	1.62	64.0	1	0	2.0	3.0	1
1	0	21.0	1.52	56.0	1	0	3.0	3.0	1
2	1	23.0	1.80	77.0	1	0	2.0	3.0	1
3	1	27.0	1.80	87.0	0	0	3.0	3.0	1
4	1	22.0	1.78	89.8	0	0	2.0	1.0	1

5 rows × 21 columns

Feature Scaling

In [33]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train_imputed)
X_test_scaled = scaler.transform(X_test_imputed)
```

Cross-Validation

In [34]:
```
from sklearn.model_selection import cross_val_score
from sklearn.svm import SVC

svm_model = SVC()

scores = cross_val_score(
    svm_model,
    X_train_scaled,
    y_train,
    cv=5,
    scoring="accuracy"
)

print(f"SVM: Mean Accuracy: {scores.mean():.3f}")
```

```
SVM: Mean Accuracy: 0.850
```

Hypeparameter Tuning for the SVC model

In [35]:
```
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV
from sklearn.preprocessing import StandardScaler

svm = SVC(random_state=42)

param_grid = {
    "C": [0.1, 1, 10, 100],
    "kernel": ["linear", "rbf", "poly"],
    "gamma": ["scale", "auto", 0.1, 1]
}

grid = GridSearchCV(
    svm,
    param_grid,
    cv=5,
    scoring="accuracy",
    n_jobs=-1
)

grid.fit(X_train_scaled, y_train)

print("Best parameters:", grid.best_params_)
print("Best CV accuracy:", grid.best_score_)
```

```
Best parameters: {'C': 10, 'gamma': 'scale', 'kernel': 'linear'}
Best CV accuracy: 0.9401930467423361
```

Final Model Evaluation

In [36]:
```
from sklearn.metrics import accuracy_score, classification_report

best_svm = grid.best_estimator_
```

```
y_pred = best_svm.predict(X_test_scaled)

final_accuracy = accuracy_score(y_test, y_pred)
print(f"Final Test Accuracy: {final_accuracy:.3f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Final Test Accuracy: 0.938

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.97      0.95        70
           1       0.98      1.00      0.99        65
           2       0.98      0.97      0.97        59
           3       0.92      0.93      0.92        58
           4       0.89      0.84      0.86        56
           5       0.88      0.88      0.88        56
           6       0.98      0.96      0.97        54

    accuracy                           0.94       418
   macro avg       0.94      0.94      0.94       418
weighted avg       0.94      0.94      0.94       418
```

Heatmap of the final output predictions

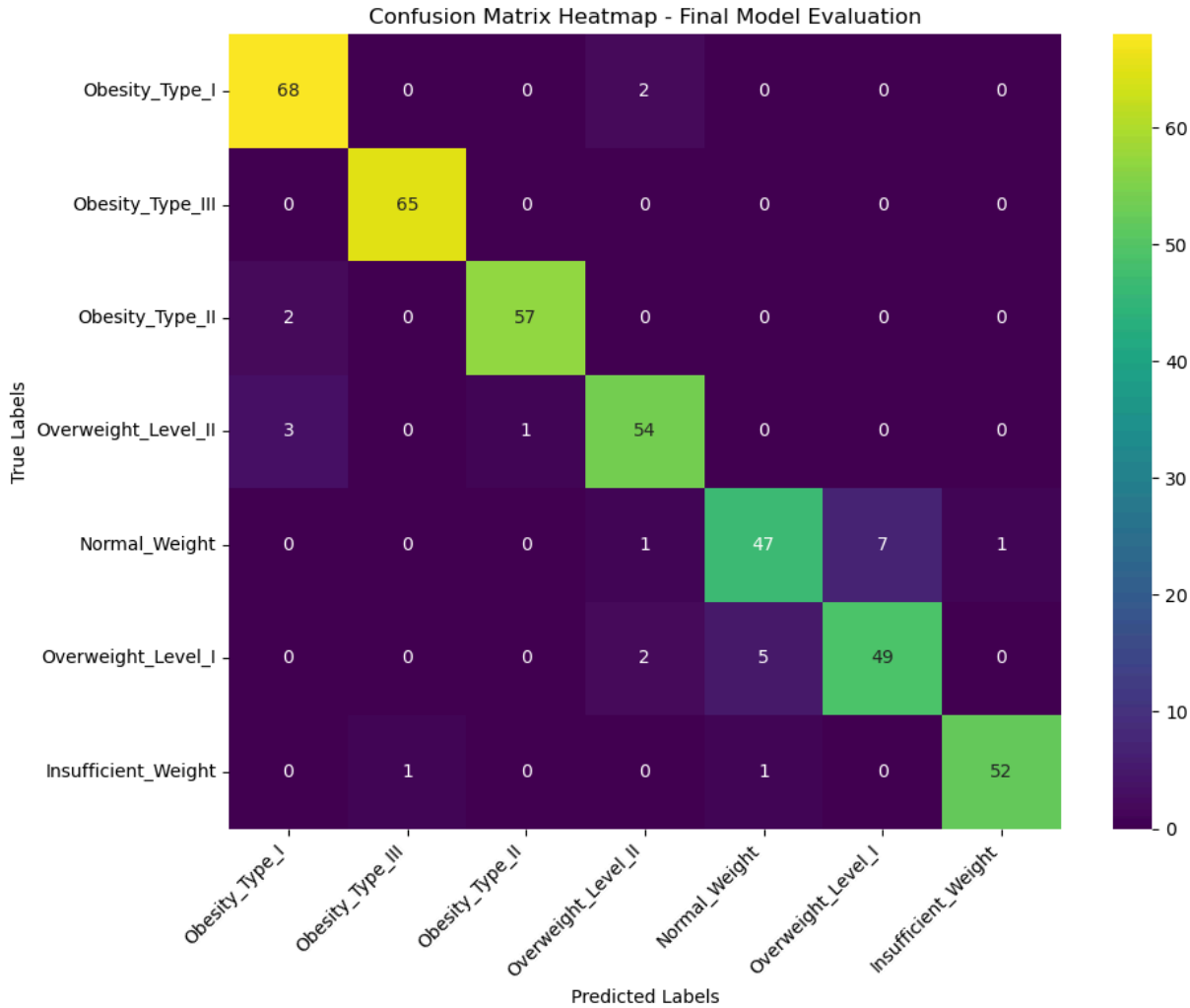
In [37]:

```
import seaborn as sns
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

labels = [
    "Obesity_Type_I",
    "Obesity_Type_III",
    "Obesity_Type_II",
    "Overweight_Level_II",
    "Normal_Weight",
    "Overweight_Level_I",
    "Insufficient_Weight"
]

plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='viridis', xticklabels=labels, yticklabel
plt.title('Confusion Matrix Heatmap - Final Model Evaluation')
plt.xlabel('Predicted Labels')
plt.ylabel('True Labels')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Save the model

In [38]:
```
import joblib

model_assets = {
    "model": best_svm,
    "scaler": scaler,
    "imputer": knn_imputer,
    "label_encoders": encoders
}

joblib.dump(model_assets, 'obesity_full_pipeline.joblib')

print("Success! Model, Scaler, Imputer, and Encoders saved to 'obesity_full_pipelin
```

```
Success! Model, Scaler, Imputer, and Encoders saved to 'obesity_full_pipeline.jobli
b'
```

In [ ]: